

Distributional invariance and proportional scaling in axonal conduction

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ABSTRACT Conduction velocity along axons reflects geometric and biophysical influences whose joint statistical organization remains largely uncharacterized. Using high-resolution time-of-arrival measurements along hundreds of identified axonal branches in vitro, we quantified how propagation speed changes along trajectories. The ratio between terminal and initial velocities, $\rho = v_{\text{end}}/v_{\text{start}}$, follows a right-skewed distribution whose shape remains invariant across branch lengths, positions within neurons, and hierarchical aggregation levels. Local conduction profiles reveal a predominantly progressive deceleration along branches. These observations indicate a simple and robust statistical organization of slowdown, suggesting that proportional modulation of propagation speed is a consistent feature of axonal signaling in structurally variable substrates.

SIGNIFICANCE Variability in axonal geometry and membrane properties is expected to influence conduction velocity, yet the statistical organization of these effects along trajectories has not been established. Using high-density microelectrode recordings from identified axonal branches, we quantify how velocity changes from origin to termination. The resulting end-to-start velocity ratio exhibits a consistent right-skewed distribution that remains stable across branch lengths, positions within neurons, and levels of aggregation. This invariance indicates that conduction slowdown reflects proportional modulation rather than idiosyncratic structural differences. The findings provide an empirical characterization of conduction dynamics in heterogeneous axonal arbors and impose clear constraints on mechanistic accounts of propagation in excitable tissue.

INTRODUCTION

Several studies of the biophysical mechanisms underlying membrane excitability have converged toward a view of the excitable membrane as a self-organizing system whose kinetics exhibit scaling relations and functional resilience across wide ranges of structural variability (1–13). These analyses, based on single-protein and whole-cell observations, suggest that reliable excitability arises from the collective kinetics of interacting conductances rather than from fixed molecular architecture. Here, we ask whether such invariant properties extend to spatially distributed settings, with conduction velocity along axons providing a simple test case.

Conduction velocity is a fundamental property of excitable media. In axons, it depends on geometry, membrane protein density, and multiple state-dependent and develop-

mental processes that generate local variations in propagation speed (13–28). Such variations can give rise to a range of mixed effects and macroscopic outcomes: if structural and kinetic factors balance geometric and branching effects, velocity remains invariant; if they do not, conduction may accelerate or slow along the trajectory. It may also vary idiosyncratically, reflecting unpredictable local contingencies. Here, we quantify the distributional form and invariance of conduction slowdown across hundreds of axonal branches, providing a unifying statistical description of how conduction velocity changes along axonal trajectories.

We report that conduction slows down systematically along axonal trajectories. The slowdown follows a statistical regularity: the ratio between end-to-start velocities is right-skewed, and its distribution remains invariant across branch lengths and positions within neurons. These regularities extend analyses of specific biophysical determinants toward a broader characterization of the stochastic processes governing axonal conduction. The distributional invariance of slowdown ratio points to a self-normalizing, proportionally

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scaling organization of conduction, in which the collective effect of many local interactions, rather than fixed structural determinants, ensures robust excitability in extended axonal trees.

MATERIALS AND METHODS

The study was conducted according to a protocol approved by the Technion, Israel Institute of Technology Committee for the Supervision of Animal Experiments (Approval IL-159-12-24).

All experiments were performed in in vitro cortical networks prepared from primary rat cortical neurons, matured in culture for 3–4 weeks (DIV 21–28) (9,29,30). Cultures were sparsely plated (initial $150 \cdot 10^3$ cells per dish; actual areal density was lower and varied due to attachment, clustering, and survival), increasing the likelihood of reconstructing well-isolated and continuous axonal branches. Recordings were obtained with CMOS-based high-density multielectrode arrays (HD-MEAs; *MaxOne*, *MaxWell* Biosystems), comprising 26,400 electrodes at a density of $3,265$ electrodes/ mm^2 and sampled at 20 kHz, a noninvasive platform enabling reconstruction of axonal trajectories from electrical activity (18,31–33). The aim of the study is not to model organism-specific brain conditions but to characterize the statistical organization of excitability in spatially extended settings; this in vitro preparation serves simply as one instance in which such invariants can be resolved.

Axonal branches were identified with the *AxonTracking* module of the *MaxLab Live* software, which links sequential electrodes exhibiting time-shifted spike-triggered averages consistent with continuous extracellular conduction along axonal processes (Fig. 1).

For each identified axonal branch, the time of arrival (TOA) at a given electrode was defined as the latency of spike detection relative to the soma. Conduction velocities at the start and end of the branch were estimated from TOA-distance relationships fitted by ordinary least squares to

the first and last 20% of sampled points with a minimum of five data points contributing to each segment fit. The end-to-start velocity ratio, $\rho = v_{\text{end}}/v_{\text{start}}$, was required to satisfy a broad plausibility range, $0 < \rho < 2.5$, chosen on the basis of velocity statistics from our laboratory's HD-MEA larger data set (mean 0.5 m/s, SD 0.12 m/s), which is consistent with published HD-MEA measurements (18,32,33). This range was deliberately broad and served only as a sanity check; all observed ratios were well within bounds, confirming that the criterion was effectively nonrestrictive.

Branches were retained only if both v_{start} and v_{end} fits achieved $R^2 \geq 0.95$. After screening, the data set consisted of 431 branches (from an initial 477) of length $L = 600\text{--}4570$ μm , recorded from 248 neurons across 14 networks. Branches excluded by the $R^2 \geq 0.95$ criterion (minimum $R^2 \approx 0.85$) had a mean slowdown ratio of $\rho \approx 0.61$, with mean fit qualities of $R^2_{\text{start}} = 0.98$ and $R^2_{\text{end}} = 0.93$, indicating (when compared with the results reported below) that exclusion reflected noisier TOA-distance relations rather than a systematic bias in slowdown.

All statistical analyses were performed in *Mathematica* and *Python* using custom scripts.

RESULTS AND DISCUSSION

Before analyzing the statistical form of ρ , we asked whether its variability might reflect differences in axonal length (which correlates strongly with the number of branch points; $r > 0.88$ (30)). Across all 431 branches, ρ showed no detectable relationship with branch length (Pearson $r = 0.018$, $p = 0.71$; Spearman $r = 0.044$, $p = 0.36$). Thus, length variation does not account for the spread of ρ .

Four examples of TOA-distance relationships are shown in Fig. 2, along with linearly fitted functions for estimation

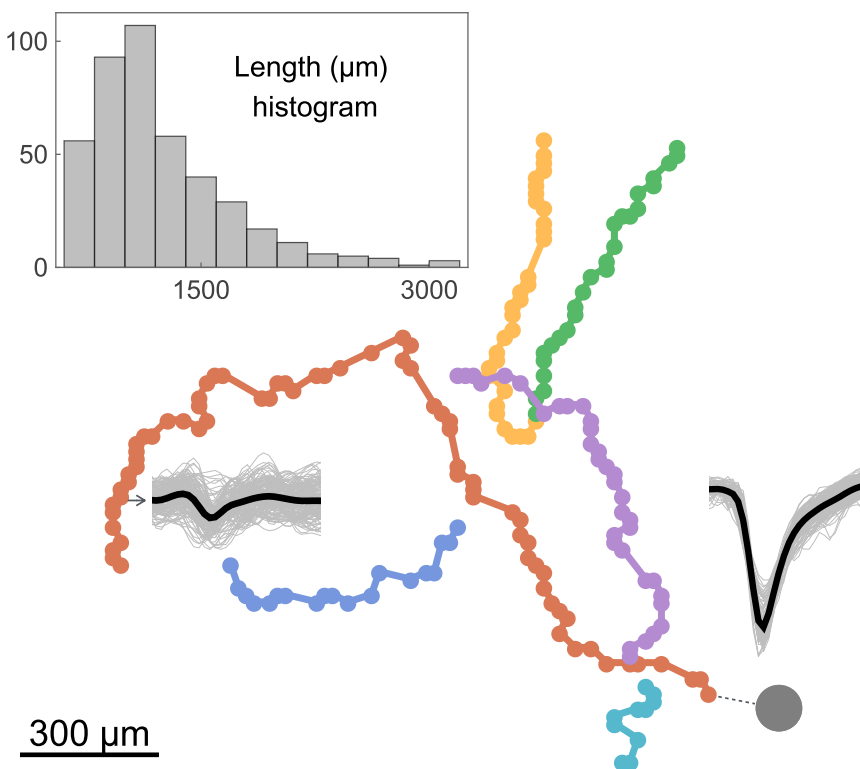


FIGURE 1 Axonal trajectories were identified using the *AxonTracking* module of *MaxLab Live*. The gray circle marks the site of somatic spike initiation, and the dashed line indicates the direction of propagation. Colored dots mark electrode positions underlying reconstructed paths. Representative voltage traces (*thin lines*, single events; *thick lines*, averages; trace duration ~ 1.5 ms) illustrate a somatic spike followed by a spike in one distal axonal electrode ~ 3.9 ms later. Scales: mean somatic spike amplitude, ~ 600 μV ; typical mean axonal spike amplitude, $4\text{--}15$ μV . The inset shows the distribution of branch lengths in the data set (431 branches from 248 neurons across 14 networks; coefficient of variation = 0.386).

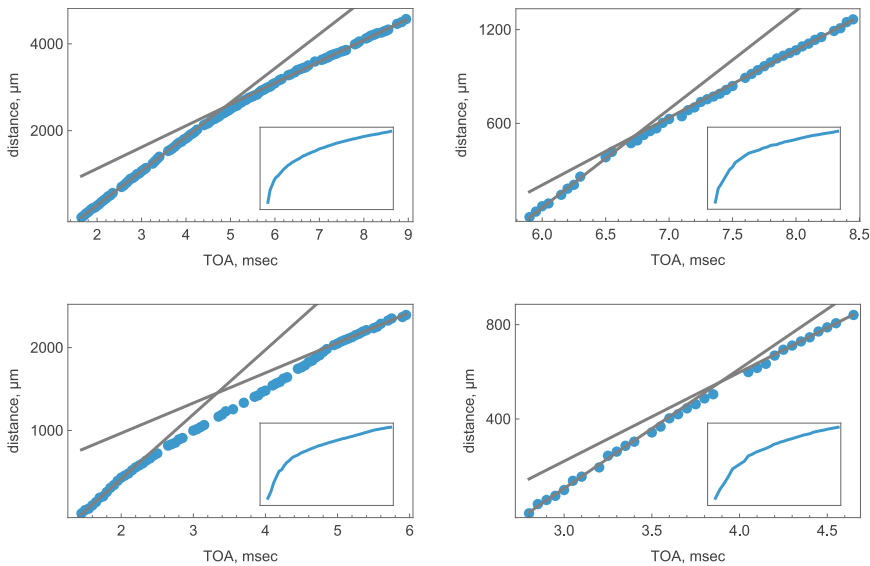


FIGURE 2 Time of arrival (TOA) versus distance for representative axonal branches, with linear fits to the initial and terminal segments used to estimate start and end conduction velocities, v_{start} and v_{end} . Insets show the same data on log-log axes, emphasizing the smooth, continuous nature of the TOA-distance relationships. The two bottom panels correspond to segments from the neuron of Fig. 1.

of end-to-start velocities, $\rho = v_{\text{end}}/v_{\text{start}}$. Log-log insets emphasize the continuous nature of the TOA-distance relationship. In all four exemplar cases of Fig. 2, a slowdown phenomenon ($\rho < 1$) is clearly visible.

The distribution of ρ across 431 branches (Fig. 3) is skewed, with typical branches slow by $\sim 30\%$ (mean $\rho \approx 0.70$, median $\rho \approx 0.68$), and a small right tail for rare acceleration (approximately 6% of branches exhibiting $\rho > 1$).

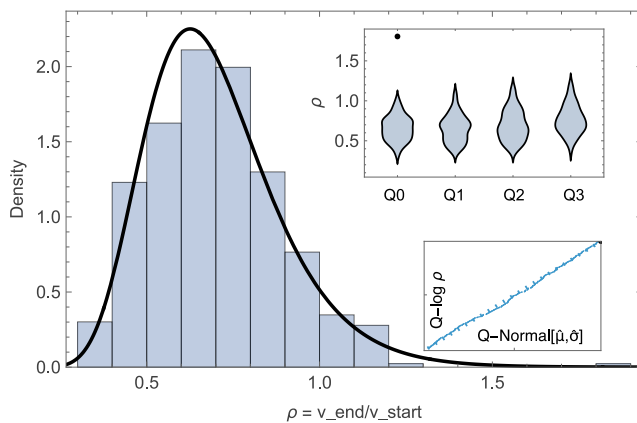


FIGURE 3 Distribution of conduction velocity ratios, $\rho = v_{\text{end}}/v_{\text{start}}$, across 431 axonal branches (coefficient of variation 0.27). The solid line shows the maximum-likelihood log-normal fit ($\hat{\mu} = -0.3943$, $\hat{\sigma} = 0.2729$; median $\rho \approx 0.68$), representative of proportional compounding models. Bottom-right inset: quantile-quantile (Q-Q) plot comparing ordered empirical $\log \rho$ values with expected quantiles of $N(\hat{\mu}, \hat{\sigma})$; the near-linear alignment indicates that $\log \rho$ closely follows a normal distribution; i.e., ρ is log-normally distributed. Top-right inset: violin representation of ρ stratified by quartiles of branch initial spike time (proxy for distance from soma). The preserved right-skewed shape across quartiles demonstrates distributional invariance; note the slight shift of the median ρ in distal (higher quartiles) branches. The black point at Q_0 reflects a single outlier observation where the calculated initial velocity was 0.35 m/s, 2.4 SD units from the average initial velocity (0.65 m/s; $n = 431$ branches).

The coefficient of variation is 0.274, significantly less compared to the coefficient of variation of branch length in the same data set, as shown in Fig. 1 ($\Delta\text{CV} = 0.11$, 95% CI [0.06, 0.16]). Were ρ merely a geometric reflection of length variability, its spread would be expected to equal or exceed that of length. The reduced variability of ρ therefore points to a normalization process in the axonal conduction machinery that compensates for geometric heterogeneity.

A maximum-likelihood estimation of the distribution shown in Fig. 3 yields $\log \rho \sim N(\hat{\mu}, \hat{\sigma})$, $\hat{\mu} = -0.3943$, $\hat{\sigma} = 0.2729$, implying median $\hat{\rho} = e^{\hat{\mu}} \approx 0.674$ and mean $\hat{\rho} \approx 0.70$, closely matching empirical values. 95% bootstrap confidence intervals (CI_{95}) were estimated for the log-normal parameters (μ , σ) and the median of ρ using 2000 resamples, yielding $\mu \in [-0.42, -0.37]$, $\sigma \in [0.26, 0.29]$, and median $\rho \in [0.66, 0.69]$. The Q-Q plot for $\log \rho$ (inset to Fig. 3) is near-linear with minor deviations.

To situate the observed log-normal form of slowdown among plausible alternatives, we conducted a series of model comparisons. We examined three representative distributional families: the *Gamma*, *log-normal*, and *log-logistic*, encompassing the main mechanistic possibilities and providing a minimal yet comprehensive basis for assessing which process best accounts for the observed form of ρ . Information criteria favored the log-normal model ($\text{AIC} = -232.18$; $\text{BIC} = -224.05$), with the Gamma distribution close behind ($\Delta\text{AIC} = \Delta\text{BIC} = 1.31$) and therefore statistically indistinguishable. However, between the log-normal and Gamma models, the former is more plausible: Gamma family represents an additive generative process, in which variability accumulates through independent increments. This is less physiologically well grounded, as excitability mechanisms, particularly sodium conductance, gating kinetics, and local impedance interactions, operate through proportional effects on current and voltage, not through additive time delays. The

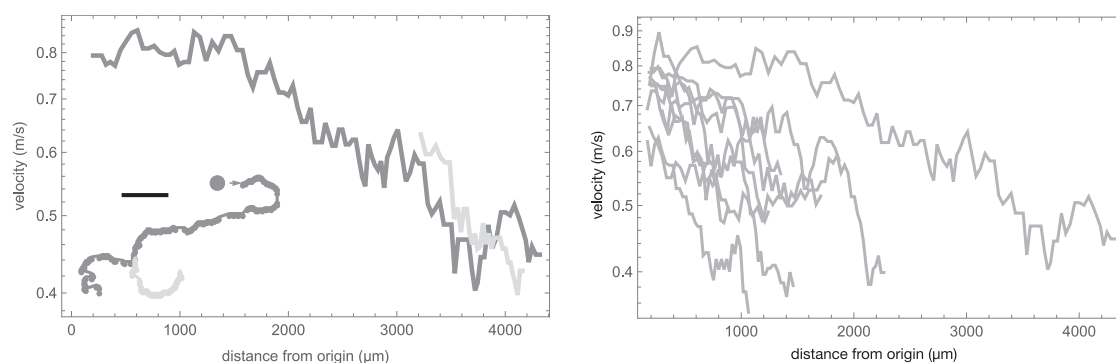


FIGURE 4 *Left*: Log-linear conduction-velocity profile along an individual bifurcating axon (moving average $\approx 350 \mu\text{m}$). The time delay between the triggering spike (its location depicted by a solid gray circle) and the first point of the long branch is $250 \mu\text{s}$. Grayscale corresponds to the branch colors shown in the lower-left inset; each branch inherits its initial velocity from the terminal velocity of its parent. Scale bar: $500 \mu\text{m}$. *Right*: Log-linear conduction-velocity profiles of axons with different lengths, all originating near the soma.

log-logistic model performed poorly ($\Delta\text{AIC} = \Delta\text{BIC} = 12.86$), providing no credible support for a regulated-multiplicative process. For completeness, we also fitted an “extreme-value” (Gumbel) distribution, which yielded $\Delta\text{AIC} = \Delta\text{BIC} = 6.36$ relative to the log-normal. Note that an extreme-value process is mechanistically difficult to reconcile with the data: The smooth convexity of TOA relations, the independence of ρ from branch length, and the near-normal distribution of $\log \rho$ all indicate that slowdown arises from the compounded effect of many small influences rather than from a bottleneck along the axon.

To examine whether hierarchical dependencies among branches, neurons, and networks influence the observed distribution of conduction slowdown, we aggregated ρ values at successive levels and re-fitted the log-normal model to each aggregate. At the neuron level, branch-specific ρ values were averaged to yield one representative value per neuron, and at the network level, neuron means were further averaged to produce one value per network. The fitted parameters remained stable across levels: branch level (reported above): $\hat{\mu} = -0.39$, $\hat{\sigma} = 0.27$; neuron level: $\hat{\mu} = -0.40$, $\hat{\sigma} = 0.24$; network level: $\hat{\mu} = -0.40$, $\hat{\sigma} = 0.13$). The corresponding medians (≈ 0.67) and means (≈ 0.69) were nearly invariant. The constancy of μ and the median indicates that the central tendency of slowdown is invariant to aggregation. The mild reduction of σ with aggregation reflects within-neuron and within-network correlations of modest magnitude. Thus, clustering exists but does not distort the log-normal form nor the overall 30% slowdown ratio.

Stratifying ρ by branch initial spike time (Fig. 3, top-right inset), a proxy for branch position relative to the soma, yielded overlapping distributions with stable shape: $\hat{\sigma} = \{0.270, 0.274, 0.276, 0.244\}$ and median $\rho = \{0.648, 0.640, 0.670, 0.743\}$ for Q0–Q3, respectively. The slight increase in ρ for branches that begin more distally is expected: conduction velocity is already lower at the onset of distal branches, reflecting the progressive deceleration evident in individual trajectories (see, for example, the left panel of Fig. 4).

As an additional diagnostic, we examined the ratio between the average and initial conduction velocities for each branch, denoted $\bar{\rho}_s = v_{\text{average}}/v_{\text{start}}$, and the ratio between the end and average conduction velocities, denoted $\bar{\rho}_e = v_{\text{end}}/v_{\text{average}}$. These measures provide a complementary view to the end-to-start ratio ($\rho = v_{\text{end}}/v_{\text{start}}$), summarizing the overall slowing along the trajectory. Across all branches, $\bar{\rho}$ exceeded ρ ($\langle \bar{\rho}_s \rangle \approx 0.88$ and $\langle \bar{\rho}_e \rangle \approx 0.79$ versus $\langle \rho \rangle \approx 0.69$). The distributions of $\bar{\rho}_s$ and $\bar{\rho}_e$ were likewise right-skewed, with approximately 18% and 10% of branches, respectively, exhibiting values greater than unity. These findings suggest that the overall deceleration is gradual and systematic. The data shown in Fig. 4 support this idea of monotonic, gradual deceleration. The left panel shows a log-linear conduction-velocity profile along one bifurcating axon (moving average $\approx 350 \mu\text{m}$). Note the gradual decline and how the branch inherits its initial velocity from the terminal velocity of its parent. Several log-linear conduction-velocity profiles of axons with different lengths, all originating near the soma, are shown in the right panel of Fig. 4. The slowdown is progressive and monotonic rather than step-like.

As shown in the left panel of Fig. 5, the histograms of v_{start} (blue) and v_{end} for 200 axons selected under a strict initial TOA criterion ($\leq 150 \mu\text{s}$) display markedly different distributional shapes. The v_{start} distribution corresponds to an average conduction velocity of 0.66 m/s , and the normal model fits the data exceptionally well: the goodness-of-fit p -value is 0.922 , the AIC is -254.26 , and the sample skewness is nearly zero (0.101), all indicating a symmetric distribution. The log-normal alternative performs worse on every metric, yielding a lower p -value (0.518), a higher AIC (-245.71), and more pronounced skewness in the log-domain (-0.541). In contrast, the v_{end} dataset exhibits the opposite pattern. Here, the log-normal model is favored, with a high p -value (0.864), the lowest AIC (-354.50), and near-zero skewness after log transformation (-0.177). The normal model, by comparison, produces a lower p -value (0.388), a substantially higher AIC

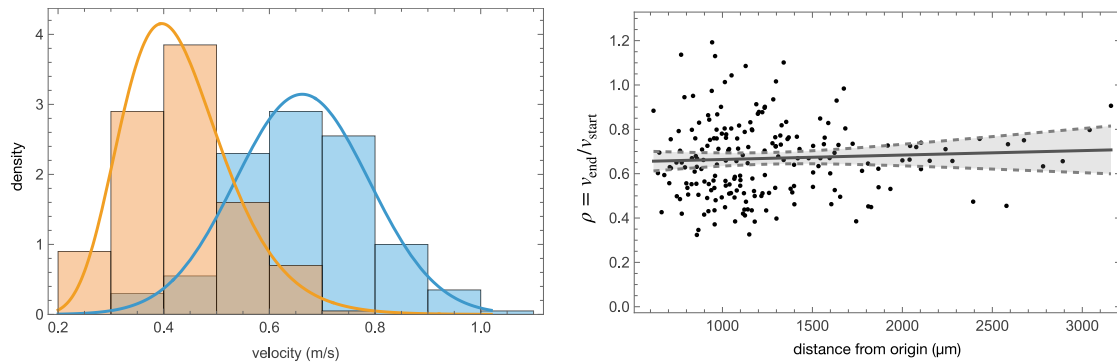


FIGURE 5 *Left:* histograms of v_{start} (blue) and v_{end} are shown for 200 axons with a strict initial time of arrival (TOA) $\leq 150 \mu s$ relative to the triggering somatic spike. For v_{start} , the average velocity (0.66 m/s) translates to branch initiation at $\leq 100 \mu m$ from the triggering spike. The normal model fits the data exceptionally well: the goodness-of-fit p -value is 0.922, the AIC is -254.26 , and the sample skewness is nearly zero (0.101), all indicating a symmetric distribution consistent with normality. The log-normal alternative performs worse on every metric, yielding a lower p -value (0.518), a higher AIC (-245.71), and more pronounced skewness in the log-domain (-0.541). In contrast, the v_{end} data set exhibits the opposite pattern. Here, the log-normal model is favored, with a high p -value (0.864), the lowest AIC (-354.50), and near-zero skewness after log transformation (-0.177). The normal model, by comparison, produces a lower p -value (0.388), a substantially higher AIC (-349.13), and a noticeably right-skewed distribution (skewness = 0.358). Taken together, these results indicate a clear transition along the axonal trajectory, from a symmetric, approximately normal distribution at the start to a skewed distribution at the end. *Right:* ρ as a function of axonal length for the 200 axons whose histograms are depicted in the left panel. The scatter is effectively random, indicating no statistically significant correlation between branch length and slowdown ratio. Shown are the linear fit and 95% confidence interval. Pearson correlation: $r = 0.05$, $p = 0.48$; Spearman rank correlation: $r_s = 0.08$, $p = 0.26$.

(-349.13), and a noticeably right-skewed distribution (skewness = 0.358). Taken together, these results indicate a clear transition along the axonal trajectory, from a symmetric, approximately normal distribution at the start to a skewed distribution at the end. Here too, there is practically no correlation between ρ and axonal length for the 200 axons whose histograms are depicted in the left panel. The scatter (right panel) is effectively random, indicating no statistically significant correlation between axonal length and slowdown ratio. Shown are the linear fit and 95% confidence interval. Pearson correlation is $r = 0.05$, $p = 0.48$; Spearman rank correlation is $r_s = 0.08$, $p = 0.26$.

Having established these empirical regularities, we now consider their relation to previous work and their mechanistic and physiological implications.

Classical analyses of tapering cables with sealed ends predict local acceleration near the distal tip due to current focusing (14). Experimental studies in cultured neurons (18,21,22,28,34) reported several forms of local velocity modulation, including slowing during spontaneous bursting, gradual developmental speeding, brief stimulus-locked acceleration, and, in some cases, reduced velocities in distal segments. These observations document important local and context-dependent effects. Here, we quantify the net, statistically invariant profile of velocity change along complete trajectories and show that it follows a simple log-normal law with length-invariant dispersion. At the level of the end-to-start ratio ρ , the various influences combine into an effective distal factor that is, on average, less than unity, yielding a proportional slowdown along the trajectory.

The approximate invariance of slowdown across branch lengths and aggregation levels suggests that propagation

speed is governed by a stochastic process in which local influences compound to produce a distribution that remains stable across branches, neurons, and networks. The essential empirical constraint is the invariance itself: any viable mechanistic account must reproduce the stability of the distribution under structural diversity. This behavior is consistent with, though not exclusive to, multiplicative dynamics, in which each axonal segment rescales velocity by a fractional factor reflecting its geometry or biophysical state, and the cumulative effect yields the observed statistical form.

Processes of this sort are familiar in transport through disordered or heterogeneous media, where local fluctuations compound multiplicatively to produce right-skewed distributions (35–38), though in those systems variance typically grows with depth, unlike the present length-invariant case. In axons, several mechanisms could instantiate such multiplicative organization: current splitting at bifurcations, impedance mismatches, gradual diameter taper, and state-dependent channel kinetics. None is mutually exclusive; together they may constitute the effective stochastic law governing propagation. A compact way to express this is to view the end-to-start ratio as a product of many small, independent or weakly dependent factors ($\rho = \prod_i k_i$), where each k_i represents a local geometric or kinetic influence. If the $\log k_i$ have finite variance, then $\log \rho$ approaches a normal distribution by the central limit theorem, yielding the observed log-normal form; empirically, this form persists even though both its mean and variance are effectively length invariant, a structural regularity that calls for theoretical analysis but lies beyond the scope of the present report. This regularity, however, imposes a clear constraint: axonal propagation cannot be captured by additive or diffusive

accumulation of variability along the trajectory. Any cable-theoretic or numerical framework must allow local influences to compound multiplicatively while incorporating a normalization mechanism that prevents dispersion from increasing with path length. Models in which variability grows monotonically with distance are therefore incompatible with the observed length-invariant slowdown.

A particularly relevant example of a potentially multiplicative mechanism is the inactivation of voltage-gated sodium channels, whose fractional availability is activity dependent over wide temporal scales (4,39). Because channel gating enters multiplicatively into total sodium conductance, variations in availability modulate conductance proportionally. Since propagation velocity scales with instantaneous sodium conductance, such proportional modulation can compensate for structural variability and support robust function. More generally, the observed invariance of the slowdown distribution suggests that global perturbations affecting excitability, such as changes in channel densities, ionic composition, temperature, or pH, would be expected to modulate distribution parameters rather than its form, a prediction that can be tested in future work.

All measurements in this study were obtained from unmyelinated axons in vitro, and the reported slowdown and its distributional invariance pertain to this conduction regime. Myelination introduces a distinct mode of propagation, and the present results do not address whether the same statistical organization would hold in myelinated fibers. If analogous proportional modulation were present, it would likely operate at different spatial scales, such as across internodes or nodes of Ranvier.

From a physiological standpoint, the robustness of conduction to structural variability is fundamental. Neural tissue operates within a geometrically irregular and dynamically mutable medium, where precise architectural reproducibility is neither attainable nor required. That axonal propagation remains statistically invariant despite such variability suggests that the organization of excitability is tuned for functional reliability rather than for geometric precision. Proportional, self-normalizing regulation ensures that local differences in geometry or conductance compound without degrading transmission fidelity. Conduction dynamics are stabilized by the same scale-invariant statistical organization that sustains excitability itself (13), enabling faithful signal transmission in a structurally heterogeneous substrate.

DATA AND CODE AVAILABILITY

Data supporting the findings of this study are available from the corresponding author upon reasonable request.

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AUTHOR CONTRIBUTIONS

L.D.C. and S.M. jointly designed the study, conducted the experiments, analyzed the data, and wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

REFERENCES

1. Millhauser, G. L., E. E. Salpeter, and R. E. Oswald. 1988. Diffusion models of ion-channel gating and the origin of power-law distributions from single-channel recording. *Proc. Natl. Acad. Sci. USA*. 85:1503–1507.
2. Teich, M. C. 1989. Fractal character of the auditory neural spike train. *IEEE Trans. Biomed. Eng.* 36:150–160.
3. Liebovitch, L., and T. Tóth. 1991. Distributions of activation energy barriers that produce stretched exponential probability distributions for the time spent in each state of the two state reaction. *Bull. Math. Biol.* 53:443–455.
4. Toib, A., V. Lyakhov, and S. Marom. 1998. Interaction between duration of activity and time course of recovery from slow inactivation in mammalian brain Na^+ channels. *J. Neurosci.* 18:1893–1903.
5. Lowen, S. B., L. S. Liebovitch, and J. A. White. 1999. Fractal ion-channel behavior generates fractal firing patterns in neuronal models. *Phys. Rev.* 59:5970–5980.
6. Ellerkmann, R. K., V. Riazanski, ..., H. Beck. 2001. Slow recovery from inactivation regulates the availability of voltage-dependent sodium channels in hippocampal granule cells, hilar neurons and basket cells. *J. Physiol.* 532:385–397.
7. Marder, E. 2011. Variability, compensation, and modulation in neurons and circuits. *Proc. Natl. Acad. Sci. USA*. 108:15542–15548.
8. Goychuk, I., and P. Hänggi. 2004. Fractional diffusion modeling of ion channel gating. *Phys. Rev.* 70:051915.
9. Gal, A., D. Eytan, ..., S. Marom. 2010. Dynamics of excitability over extended timescales in cultured cortical neurons. *J. Neurosci.* 30:16332–16342.
10. Gal, A., and S. Marom. 2013. Self-organized criticality in single-neuron excitability. *Phys. Rev.* 88:062717.
11. Meisel, C., A. Klaus, ..., D. Plenz. 2015. Critical slowing down governs the transition to neuron spiking. *PLoS Comput. Biol.* 11:e1004097.
12. Marom, S. 2016. Emergence and maintenance of excitability: kinetics over structure. *Curr. Opin. Neurobiol.* 40:66–71.
13. Marom, S., and E. Marder. 2023. A biophysical perspective on the resilience of neuronal excitability across timescales. *Nat. Rev. Neurosci.* 24:640–652.
14. Goldstein, S. S., and W. Rall. 1974. Changes of action potential shape and velocity for changing core conductor geometry. *Biophys. J.* 14:731–757.
15. Segev, I., and E. Schneidman. 1999. Axons as computing devices: basic insights gained from models. *J. Physiol. Paris*. 93:263–270.
16. Bucher, D., and J.-M. Goaillard. 2011. Beyond faithful conduction: short-term dynamics, neuromodulation, and long-term regulation of spike propagation in the axon. *Prog. Neurobiol.* 94:307–346.
17. Debanne, D., E. Campanac, ..., G. Alcaraz. 2011. Axon physiology. *Physiol. Rev.* 91:555–602.

18. Bakkum, D. J., U. Frey, ..., A. Hierlemann. 2013. Tracking axonal action potential propagation on a high-density microelectrode array across hundreds of sites. *Nat. Commun.* 4:2181.
19. Seidl, A. H. 2014. Regulation of conduction time along axons. *Neuroscience*. 276:126–134.
20. El Hady, A., and B. B. Machta. 2015. Mechanical surface waves accompany action potential propagation. *Nat. Commun.* 6:6697.
21. Shimba, K., K. Sakai, ..., Y. Jimbo. 2015. Axonal conduction slowing induced by spontaneous bursting activity in cortical neurons cultured in a microtunnel device. *Integr. Biol.* 7:64–72.
22. Habibey, R., S. Latifi, ..., A. Blau. 2017. A multielectrode array micro-channel platform reveals both transient and slow changes in axonal conduction velocity. *Sci. Rep.* 7:8558.
23. Rama, S., M. Zbili, and D. Debanne. 2018. Signal propagation along the axon. *Curr. Opin. Neurobiol.* 51:37–44.
24. Alcamí, P., and A. El Hady. 2019. Axonal computations. *Front. Cell. Neurosci.* 13:413.
25. Emmenegger, V., M. E. J. Obien, ..., A. Hierlemann. 2019. Technologies to study action potential propagation with a focus on HD-MEAs. *Front. Cell. Neurosci.* 13:159.
26. Kamiya, H., and D. Debanne. 2020. Axon neurobiology: fine-scale dynamics of microstructure and function. *Front. Cell. Neurosci.* 14.
27. Zang, Y., E. Marder, and S. Marom. 2023. Sodium channel slow inactivation normalizes firing in axons with uneven conductance distributions. *Curr. Biol.* 33:1818–1824.e3.
28. Tamatani, C., K. Shimba, ..., Y. Jimbo. 2023. Activity- and spatial-dependent variations in axonal conduction recorded from microtunnel electrodes. In 2023 15th Biomedical Engineering International Conference (BMEiCON) IEEE, pp. 1–3.
29. Eytan, D., and S. Marom. 2006. Dynamics and effective topology underlying synchronization in networks of cortical neurons. *J. Neurosci.* 26:8465–8476.
30. Cohen, L. D., T. Galateanu, and S. Marom. 2026. Dynamics of excitability in axonal trees. *Biophys. J.* 125:245–254.
31. Müller, J., M. Ballini, ..., A. Hierlemann. 2015. High-resolution CMOS MEA platform to study neurons at subcellular, cellular, and network levels. *Lab Chip*. 15:2767–2780.
32. Radivojevic, M., F. Franke, ..., D. J. Bakkum. 2017. Tracking individual action potentials throughout mammalian axonal arbors. *eLife*. 6:e30198.
33. Radivojevic, M., and A. Rostedt Punga. 2023. Functional imaging of conduction dynamics in cortical and spinal axons. *eLife*. 12:e86512.
34. Lewandowska, M. K., M. Radivojević, ..., A. R. Hierlemann. 2016. Cortical axons, isolated in channels, display activity-dependent signal modulation as a result of targeted stimulation. *Front. Neurosci.* 10:83.
35. Landauer, R. 1970. Electrical resistance of disordered one-dimensional lattices. *Philos. Mag.* 21:863–867.
36. Kirkpatrick, S. 1973. Percolation and conduction. *Rev. Mod. Phys.* 45:574–588.
37. Bouchaud, J.-P., and A. Georges. 1990. Anomalous diffusion in disordered media: statistical mechanisms, models and physical applications. *Phys. Rep.* 195:127–293.
38. Limpert, E., W. A. Stahel, and M. Abbt. 2001. Log-normal distributions across the sciences: keys and clues. *Bioscience*. 51:341–352.
39. Ulbricht, W. 2005. Sodium channel inactivation: molecular determinants and modulation. *Physiol. Rev.* 85:1271–1301.